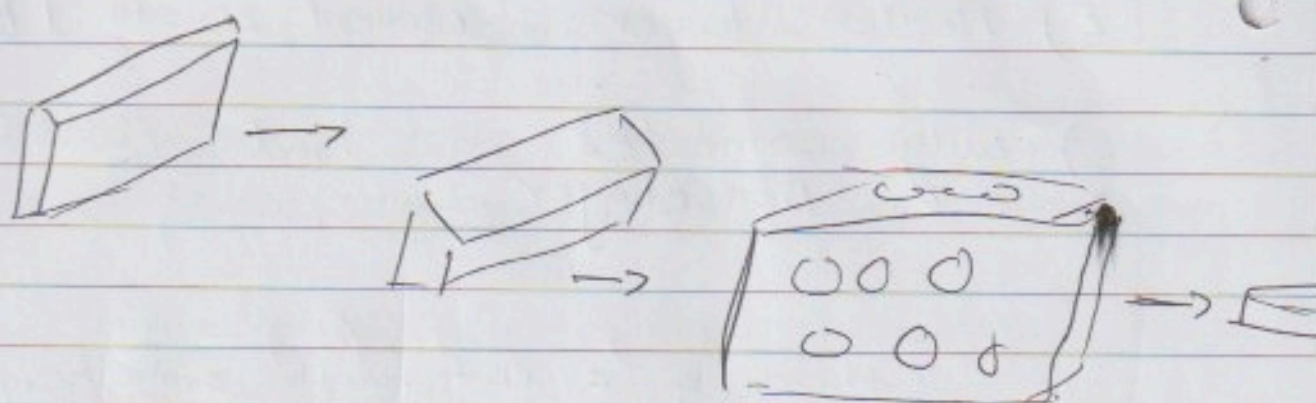




CNN

Arranges neuron in

3D width, height, length

Listen
G:41

Input $\rightarrow$ Convolution $\rightarrow$ Relu - Pool - FC.

What is Kernel.

Pixel.

This is not Matrix multiplication

5x5

$$\begin{bmatrix} 1 & 2 & 3 & 0 & 1 \\ 0 & 1 & 2 & 3 & 1 \\ 1 & 2 & 1 & 0 & 1 \\ 2 & 1 & 0 & 1 & 2 \\ 1 & 0 & 2 & 3 & 1 \end{bmatrix}$$

$$\times \begin{bmatrix} 1 & 0 & -1 \\ 1 & 0 & -1 \\ 1 & 0 & -1 \end{bmatrix}$$

This kernel identifies edges.

Moving pixel is called stride.

$$2 + 0 + 0 + 1 + 0 - 3$$

$$= 1 + 0 = 3 + 0 + 0 - 2 + 1 + 0 - 1$$

$$= -4$$

Kernel purpose is to help algo by extra features. Kernels are also called filters.
List of kernels in numpy.